CVR COLLEGE OF ENGINEERING

(An UGC Autonomous Institution with NAAC 'A' Grade,Affiliated to JNTUH)
Vastunagar, Mangalpalli (V), Ibrahimpatan (M), R.R.District, Telangana
Ph.No:91-8414 – 661601, 661675

DEPARTMENT OF CSE (DATA SCIENCE)

COURSE OUTCOMES (COs)
OF
B. TECH. **CSE(DATA SCIENCE)**

R22 REGULATIONS

I Year I Semester			
Course Code	**Course Title**	**CO Number**	**Course Outcome Statement**
22BS101/151	**Applied Physics**	CO1	Understand various aspects of Lasers and Optical fiber and their applications in diverse fields.
		CO2	Understand physical world from fundamental point of view by the concepts of Quantum mechanics and visualize the difference between conductor, semiconductor, and an insulator by classification of solids.
		CO3	Identify the role of semiconduc
		CO4	tor devices in science and engineering Applications.
		CO5	Explore the fundamental properties of dielectric, magnetic and energy materials, their applications.
22BS103	**Mathematics for Computing**	CO1	Solve the first-order Ordinary Differential Equations and extend the knowledge to the applications in engineering problems.
		CO2	Solve higher-order Ordinary Differential Equations and Difference equations and extend the knowledge to the applications in engineering problems.
		CO3	Find the extremum of a multi-variate function with/without constraints.
		CO4	Determine the convergence/divergence of a given infinite series.
		CO5	Solve the first-order Ordinary Differential Equations and extend the knowledge to the applications in engineering problems.
22HS101/151	**English For Skill Enhancement**	CO1	Choose appropriate vocabulary and sentence structures for oral and written communication suitable to the context and culture.
		CO2	Communicate effectively in various professional contexts through oral and written communication.
		CO3	Comprehend, emphasize, conceptualize and evaluate the given texts and other authentic texts such as magazines, newspaper articles etc.
		CO4	Understand explicit and implicit meaning and draw inference from the given text.
		CO5	Evaluate their language skills and soft skills to handle personal and Professional challenges.
22CS101	**Problem Solving Through C**	CO1	Understand programming concepts and analyze a problem, design a solution and develop an algorithm to solve it.
		CO2	Modularize a problem and implement the solution using basic programming concepts, control statements and functions.

		CO3	Evaluate the use of macros and implement solutions to complex problems using recursion and homogeneous data types.
		CO4	Implement solution using pointers for problems of relevance and use different dynamic memory allocation methods.
		CO5	Understand and analyze, differentiate and implement elementary algorithms of sorting, searching and will also be able to compare and contrast algorithms with respect to time and space complexity.
22CS102	**Essentials of System and Web Interfacing**	CO1	Interact with the UNIX system modules and utilize utilities
		CO2	Apply file filters and generate analytics using awk and grep tools
		CO3	Develop efficient shell scripts using bash constructs.
		CO4	Develop a simple web page using HTML elements.
		CO5	Present the content with good user experiences and make pages responsive.
22HS102/152	**Environmental Science**	CO1	Define the concepts of Ecosystem.
		CO2	Explain and enunciate the value of biodiversity and its conservation.
		CO3	Discuss various natural resources and their importance, understand the advantages and disadvantages of renewable energy sources and technologies.
		CO4	Develop awareness on pollution control technologies and global atmospheric Changes.
		CO5	Relate the importance of Environmental Impact Assessment and Environmental legislation in the management of the environment.
22BS131/181	**APPLIED PHYSICS LAB**	CO1	Determine the wavelength of light by diffraction principle and learn methods to minimize the signal loss in optical fibers
		CO2	Understand the applications of various semiconductors and optoelectronic devices.
		CO3	Gain knowledge of applications of dielectric materials and hysteresis behavior of magnetic materials.
		CO4	Understands the concepts of resonance, charging and discharging of the capacitor, dispersion of light and carry out data analysis.
		CO5	Gain knowledge of applications of dielectric materials and hysteresis behavior of magnetic materials.
22CS131	**C PROGRAMMING LAB**	CO1	Familiarize with Linux programming environment and translate given algorithms to a working and correct program.
		CO2	Interpret syntax errors as reported by the compilers and to be able to identify and correct logical errors encountered at run time using debuggers like GDB.
		CO3	Write iterative as well as recursive programs.
		CO4	Represent data in arrays, pointers, strings and manipulate them through a program
		CO5	Apply Algorithm for solving problems like sorting, searching.
22ME131/181	**ENGINEERING WORKSHOP**	CO1	Acquire skills of basic engineering trades like Carpentry, Tin smithy etc.
		CO2	Demonstrate an understanding of and comply with workshop safety regulations
		CO3	Identify and use marking out tools, hand tools, measuring equipment and to work to prescribed tolerances.
		CO4	Apply the knowledge of the above trades in their day –to – day activities.

		CO5	Select appropriate equipment and consumables for required applications.
22CS132	**ESSENTIALS OF SYSTEM AND WEB INTERFACING LAB**	CO1	Navigate through the Linux file system and specify access permission on new or existing files.
		CO2	Express and implement pattern-matching techniques using grep.
		CO3	Design blogs and beautiful web pages using HML and CSS.
		CO4	Format word documents with various contents such as tables, and figures, extract data analytics from excel file and present the insights using PowerPoint.
		CO5	Navigate through the Linux file system and specify access permission on new or existing files.

I Year II Semester			
Course Code	**Course Title**	**CO Number**	**Course Outcome Statement**
22BS152/102	**ENGINEERING CHEMISTRY**	CO1	Relate the basic properties of water and its usage for domestic and industrial purposes.
		CO2	Summarize the basic knowledge of electrochemical procedures related to batteries and corrosion and its control.
		CO3	Apply the fundamentals and general properties of polymers and other engineering materials.
		CO4	Analyze real-time situations related fuel energy sources.
		CO5	Predict potential applications of chemistry and the practical utility of engineering materials in order to become good engineers and entrepreneurs.
22BS154	**APPLIED LINEAR ALGEBRA**	CO1	Model high dimensional data using matrices and evaluate the rank of matrices.
		CO2	Evaluate Eigenvalues, and Eigenvectors, and find the Modal matrix under a linear transformation.
		CO3	Apply the concept of matrix derivatives in various machine learning techniques.
		CO4	Appreciate the concept of vector spaces and solve real-world engineering problems using Least squares approximations.
		CO5	Demonstrate the decomposition techniques of the matrix to optimize the computational complexity.
22CS151	**: DATA STRUCTURES THROUGH C**	CO1	Design and implement appropriate user-defined types to a given problem definition and apply various functions for processing files.
		CO2	Understand basic concepts, Design and implement linear list data structures.
		CO3	Implement stack and queue data structures and their application.
		CO4	Assimilate the terminology of trees and implement binary tree operations in C.
		CO5	Understand the representation of graph and traversal techniques.
22ME151/101	**COMPUTER AIDED ENGINEERING DRAWING**	CO1	Know the Standard conventions and Construction of various Engineering curves through Auto CAD.
		CO2	Apply fundamentals of theory of projections and draw orthographic projections of points and lines in any position through Auto CAD.
		CO3	Construct orthographic projections of simple planes and regular solids in any position through Auto CAD.
		CO4	Draw sectional views and developments of various basic 3D objects through Auto CAD.

		CO5	Construct isometric views and construct multi view drawings of simple 3D objects through Auto CAD.
22IT151	**PYTHON FOR COMPUTING**	CO1	Write programs using various control structures of Python.
		CO2	Use functions and modules for problem-solving.
		CO3	Design and develop solutions to real-world problems using available data structures in Python Language.
		CO4	Write programs using files & Advanced features of Python.
		CO5	Know the importance of object-oriented programming in Python & analyze data with Numpy
22BS132/182	**ENGINEERING CHEMISTRY LAB**	CO1	Determine the parameters like the hardness of water, alkalinity, and rate of corrosion of mild steel
		CO2	Estimate the acid concentration by conductometry
		CO3	Analyze instrumental techniques such as potentiometry and pH meter in order to find out the concentrations or equivalence points.
		CO4	Interpret molecular/system properties such as viscosity, and saponification value of coconut oil.
		CO5	Apply analytical skills about colorimeter/ polymer/Sanitizer.
22HS181/131	**ENGLISH LANGUAGE AND COMMUNICATION SKILLS LAB**	CO1	Comprehend and respond appropriately in various scenarios.
		CO2	Emerge as confident and competent communicators of the English Language.
		CO3	Apply pronunciation skills to evolve as proficient speakers.
		CO4	Analyze and compose effectively across various mediums.
		CO5	Develop critical and analytical thinking.
22CS181	**DATA STRUCTURES THROUGH C LAB**	CO1	Implement file processing functions and be able to store, retrieve and process data in text and binary format
		CO2	Understand basic data structures such as arrays, and linked lists.
		CO3	Understand basic data structures such as stacks, queues, and circular Queues
		CO4	Implement operations on Binary Search Trees
		CO5	Solve problems involving graphs.
22IT181	**PYTHON FOR COMPUTING LAB**	CO1	Write programs using various control structures of Python.
		CO2	Use functions and modules for problem-solving.
		CO3	Design and develop solutions to real-world problems using available data structures in Python Language.
		CO4	Write programs using files & Advanced features of Python.
		CO5	Know the importance of object-oriented programming in Python & analyze data with NumPy.

III Semester			
Course Code	**Course Title**	**CO Number**	**Course Outcome Statement**
22CS202	**OBJECT ORIENTED PROGRAMMING THROUGH JAVA**	CO1	Design and implement object-oriented concepts like encapsulation, abstraction and data hiding using programming constructs offered by java language.
		CO2	Realize the power of inheritance, interfaces, and packages.
		CO3	Understand and demonstrate the concepts of exception handling and java io streams.
		CO4	Demonstrate knowledge and understanding of multi-threading, annotations, and modules in Java.
		CO5	Design and develop java applications using AWT & Swings and make use of the advanced features for providing solutions to real world problems.
22CS201	**DISCRETE MATHEMATICS**	CO1	Apply formal logic proofs and/or informal, but rigorous, logical reasoning to evolve theoretical proofs to real problems, such as predicting the behavior of software or solving problems such as puzzles.
		CO2	Apply the logical notations to define and reason about fundamental mathematical concepts such as sets, and relations and exercise the guidelines for constructing valid arguments. A representation of a partially ordered set such as a lattice as a directed graph.
		CO3	Define Group properties and construct simple functions that preserve the algebraic structures over groups.
		CO4	Solve counting problems efficiently by applying the principle of inclusion and exclusion and solve recurrence relations.
		CO5	Characterize edge preserving similarity between two graphs and verify the Eulerian property of graphs.
22HS204	**MATHEMATICAL AND STATISTICAL FOUNDATIONS**	CO1	Apply the Number Theory concepts to the cryptography domain
		CO2	Compute probabilities using theorems in probability and probability distributions
		CO3	Find estimates of parameters and test hypothesis about parameters
		CO4	Establish a relationship between variables using correlation and regression
		CO5	Take decisions using testing of hypothesis techniques & Analyze live data
		CO6	Resolve the potential misconceptions and hazards in each topic of study
22CS203/253	**DATABASE MANAGEMENT SYSTEMS**	CO1	Demonstrate an understanding of database management system components and features. Design E-R Model to represent real-world database application scenarios.
		CO2	Demonstrate mathematical approach towards querying database using relational algebra and relational calculus and implement using SQL.
		CO3	Convert E-R Model to a relational Model and design proper relational database while eliminating anomalies.
		CO4	Demonstrate the role of transaction management and concurrency control protocols.
		CO5	Demonstrate an understanding of database file organization and recovery of the database in case of crashes.
22DT201	**DIGITAL ELECTRONICS AND DESIGN**	CO1	Master different number systems and realize the binary operations of Boolean algebra using logic gates.
		CO2	Simplification of Complex problems of gate-level circuits using K-Map and simple circuit implementations.

		CO3	Student has the knowledge about the design and functionalities of Combinational circuits.
		CO4	Design of Synchronous Sequential circuits and its working principles.
		CO5	Explain the concepts of basic memory and programmable array logics used in a digital system.
22CS231	**OBJECT ORIENTED PROGRAMMING THROUGH JAVA LAB**	CO1	Implement object-oriented concepts like encapsulation, data hiding and abstraction using programming constructs offered by java language.
		CO2	Develop java programs to realize the power of inheritance, interfaces and packages.
		CO3	Develop java programs to demonstrate the concepts of exception handling and I/O streams.
		CO4	Implement java applications using multithreading mechanism and understand the power of modules.
		CO5	Use graphical user interfaces to create Frames for providing solutions to real world problems.
22CS232/282	**DATABASE MANAGEMENT SYSTEMS LAB**	CO1	Analyze database requirements and determine the entities involved in the system and their relationship to each other.
		CO2	Design E-R Model to represent database application scenarios.
		CO3	Convert/transform the E-R Model to relational tables, populate relational database and formulate SQL queries on data.
		CO4	Improve the database design by normalization.
		CO5	Implement PL/SQL procedures, function, triggers and cursors.
22CS232/282	**DATABASE MANAGEMENT SYSTEMS LAB**	CO1	Analyze database requirements and determine the entities involved in the system and their relationship to each other.
		CO2	Design E-R Model to represent database application scenarios.
		CO3	Convert/transform the E-R Model to relational tables, populate relational database and formulate SQL queries on data.
		CO4	Improve the database design by normalization.
		CO5	Implement PL/SQL procedures, function, triggers and cursors.
22DT231	**DATA VISUALIZATION LAB**	CO1	Identify the different data types, visualization types to bring out the insight.
		CO2	Relate the visualization towards the problem based on the dataset to analyze and bring out valuable insight on a large dataset.
		CO3	Demonstrate the analysis of a large dataset using various visualization techniques and tools.
		CO4	Identify the different attributes and showcasing them in plots. Identify and create various visualizations for geospatial and table data.
		CO5	Ability to create and interpret plots using R/Python.
22HS201/251	**CONSTITUTION OF INDIA**	CO1	Understand and explain the significance of Indian Constitution as the fundamental law of the land.
		CO2	Discuss the intellectual origins of the framework of argument that informed the conceptualization of social reforms leading to revolution in India. Exercise his fundamental rights in proper sense at the same time identifies his responsibilities in national building.
		CO3	Analyze the organs of governance and District's Administration head

		CO4	Analyze the Local Administration: District's and Village Administration
		CO5	Understand Election Commission Process and Institutional Bodies for the welfare of SC/ST/OBC and women.

IV Semester			
Course Code	**Course Title**	**CO Number**	**Course Outcome Statement**
22CS252	**ADVANCED DATA STRUCTURES THROUGH JAVA**	CO1	Realize the power of generics and functional programming in java.
		CO2	Understand Java's Collection class hierarchy and also know the power of data processing using streams.
		CO3	Implement dictionaries using linear lists, hashing & binary search tree and compare their performances.
		CO4	Implement dictionaries using an AVL tree and red, black tree.
		CO5	Understand the advantages of B-trees and Priority Queues.
22DT251	**DESIGN AND ANALYSIS OF ALGORITHMS**	CO1	Analyze worst-case running times using asymptotic analysis of algorithms.
		CO2	Describe the divide-and-conquer paradigm and Synthesize divide-and-conquer algorithms.
		CO3	Define optimization problems and solve them through various greedy policies
		CO4	Describe the dynamic-programming paradigm and synthesize dynamic programming algorithms and analyze them.
		CO5	Reduce the size of search space of the optimization problems by applying backtracking and branch and bound tools. Appreciate the Non-Deterministic modeling of algorithms.
22CY251	**COMPUTER ORGANIZATION AND ARCHITECTURE**	CO1	Fundamentals of Digital Computer Organization and Architecture, its basic concepts and functions, pipeline and parallel computing.
		CO2	Understand the deep operations of Registers & Memory with notation language using RTL and Micro operations. Instruction formats with respect to computer system.
		CO3	The execution of Micro program from control memory and how the hardwired control is designed through combinational logic and detailed CPU organization, Basics of RISC & CISC.
		CO4	Data representations as per IEEE standards, various arithmetic operations and various algorithms for arithmetic operations multiplication and division.
		CO5	Importance of memory, different types of memory and its operations, I-O operations and Multiprocessor Architecture.
22AM251	**SOFTWARE ENGINEERING AND DEVOPS**	CO1	Master underlying principles of software engineering, and Evolution of various Software Process models while analyzing the agile principles, practices.
		CO2	Apply Requirements planning and implement best practices in software development.
		CO3	Analyze continuous Integration and Continuous Deployment in Industrial Scenario.
		CO4	Apply various Tool Suits available for implementation of DevOps.
		CO5	Apply Tools used in DevOps to Cloud based Applications while understanding the future trends and current best practices.

22IT252	**OPERATING SYSTEMS**	CO1	Understand the role of Operating System with its function and services.
		CO2	Compare various algorithms used for CPU scheduling and apply various concepts related to concurrency and synchronization to solve problems.
		CO3	Understand the inter process communication mechanism and resolve deadlock in a multi-programmed environment.
		CO4	Understand the concepts of virtual memory and how it is realized in systems
		CO5	Differentiate and Demonstrate file systems, directory structures and their implementation issues.
22CS281	**ADVANCED DATA STRUCTURES THROUGH JAVA LAB**	CO1	Understand the power of generics and functional programming.
		CO2	Implement hashing, sets, stacks and queues using collection classes in java.util package and process the data using streams.
		CO3	Implement dictionaries using various data structures like sorted list and hashing.
		CO4	Implement dictionaries using various height-balanced trees and also analyze the advantages and disadvantages of height-balanced trees.
		CO5	Understand the importance of Priority Queues and their applications.
22AM281	**SOFTWARE ENGINEERING AND DEVOPS LAB**	CO1	Implement CASE tools for software design.
		CO2	Write test cases & automate them for execution using J Unit
		CO3	Use Git and Jenkins for Version Control and Continuous Integration
		CO4	Apply Maven and ANSIBLE for Build Automation and implement integration with Jenkins and Docker
		CO5	
22IT282	**OPERATING SYSTEMS LAB**	CO1	Simulate and implement operating system concepts such as scheduling, deadlock management, file management and memory management.
		CO2	Implement C programs using Unix system calls.
		CO3	
		CO4	
		CO5	
22DT284	**Real-Time/Field-Based Research Project**	CO1	Identify and formulate a real-world data-driven problem through comprehensive literature review.
		CO2	Design a structured methodology for data acquisition, cleaning, exploration, and hypothesis formulation.
		CO3	Apply appropriate data science models, machine learning algorithms, and tools to extract patterns and build predictive/analytical systems.
		CO4	Interpret results using relevant statistical, computational, and visualization techniques to derive actionable insights.
		CO5	Demonstrate research ethics, teamwork, project planning, and effective communication through technical documentation and oral presentation of project outcomes.
22HS281/231	**GENDER SENSITIZATION LAB**	CO1	Students will have developed a better understanding of important issues related to gender in contemporary India.
		CO2	Students will be sensitized to basic dimensions of the biological, sociological, psychological and legal aspects of gender.
		CO3	Students will attain a finer grasp of the biological spheres of gender in our society and how to counter it.

		CO4	Students will acquire insight into the gendered division of labor and its relation to politics and Economics.
		CO5	Students will develop a sense of appreciation for women in all walks of life and contribute to establish an egalitarian society.

V Semester			
Course Code	**Course Title**	**CO Number**	**Course Outcome Statement**
22CY304/252	**Computer Networks**	CO1	Outline the basic concepts of data communications including the key aspects of networking and their interrelationship, packet and circuit switching as internal and external operations, physical structures, types, models, and internetworking
		CO2	Make use of different types of bit errors and the concept of bit redundancy for error detection and error correction.
		CO3	Identify the suitable design parameters and algorithms for assuring quality of service and internetworking in various internet protocols
		CO4	Interpret transport protocols (TCP, UDP) services and functionalities for measuring the 1network performance
		CO5	Illustrate the various application layer protocols (FTP, POP, and SMNP), E-Mail, WWW and standards (DNS) in data communications among network
22DT301	**Introduction to Data Analysis**	CO1	Understand basic terms of data and its types
		CO2	Identify probability distributions to fit a model
		CO3	Describe the data using various statistical measures
		CO4	Utilize R elements for data handling
		CO5	Perform data reduction and apply visualization techniques
22IT301	**Web Technologies**	CO1	Build a custom website with HTML, CSS, and Bootstrap
		CO2	Demonstrate JavaScript and its asynchronous nature of execution
		CO3	Implement the Database Connectivity and Component Technologies like Beans
		CO4	Develop and deploy Servlet based web applications
		CO5	Develop Server-side programming using JSP
22IT305/254	**Introduction to IoT**	CO1	Know characteristics and physical design of IOT.
		CO2	Analyze networking and interoperability in IOT.
		CO3	Program and configure Arduino boards for various designs.
		CO4	Understand Python programming and interfacing for Raspberry Pi.
		CO5	Explore IoT applications in different domains.
22CS309/354	**Automata Theory and Compiler Design**	CO1	Convert NFA to DFA and regular expression to DFA.
		CO2	Design top-down and bottom-up parsers.
		CO3	Understand the concepts of type checking and intermediate code generation forms.
		CO4	Understand the concepts of storage allocation strategies, symbol table management and hashing.
		CO5	Use DAG for generating assembly language code and able to generate relocatable machine code.
22IT331	**Web Technologies Lab**	CO1	To build a custom website with HTML, CSS, and Bootstrap

		CO2	Demonstrate JavaScript, XML, DHTML and related Technologies
		CO3	Implement the Database Connectivity and Component Technologies like Beans.
		CO4	Deploy the servlet technology & API Management.
		CO5	Construct the fundamentals of JSP.
22DT331	**Computer Networks and Internet of Things Lab**	CO1	Implement data link layer framing methods
		CO2	Analyse error detection and error correction codes
		CO3	Implement and analyse routing and congestion issues in network design
		CO4	Use sensors to measure temperature, humidity, light, and distance
		CO5	Deploy an IoT application and connect to the cloud
22DT332	**R Programming and Compiler Design Lab**	CO1	Setup R Programming Environment
		CO2	Understand and use R – Data types and R – Data Structures
		CO3	Develop programming logic using R – Packages and Analyse data sets using R – programming capabilities
		CO4	To implement Lexical Analyzer using Lex tool & Syntax Analyzer or parser using YACC Tool
		CO5	To implement code optimization techniques
22CY333	**Advanced Web Programming Lab**	CO1	Design and build robust REST APIs using Node.js, demonstrate the Express Framework
		CO2	Understand components in React
		CO3	Handle event in React and bind event listeners
		CO4	Work with React forms
		CO5	Create applications for consuming REST APIs
22HS302/ 352/253	**Intellectual Property Rights**	CO1	Understand concepts of intellectual property rights.
		CO2	Evaluate trademark
		CO3	File for a patent
		CO4	Analyze the fairness in a competition
		CO5	Understand laws related to intellectual property rights.

VI Semester			
Course Code	**Course Title**	**CO Number**	**Course Outcome Statement**
22DT351	**Predictive Analytics**	CO1	Explore prediction-related principles, theories and approaches.
		CO2	Learn model assessment and validation.
		CO3	Use of the basics of predictive techniques and statistical approaches.
		CO4	Apply of supervised algorithms for classification.
		CO5	Analyse the data using clustering algorithms.
22AM356/301	**Machine Learning**	CO1	Formalize a real-world problem as a Machine Learning task
		CO2	Implement predictors using regression models and assess their quality

		CO3	Develop multiple-class classifiers and develop recommender systems
		CO4	Efficiently implement clustering algorithms and assess their performance
		CO5	Implement simple Q-Learning algorithm, based on Value iteration, and appreciate actor- critic model.
22DT352	**Big Data Analytics**	CO1	Describe big data and use cases from selected business domains, Learn Hadoop and HDFS concepts
		CO2	Perform map-reduce analytics using Hadoop
		CO3	Use Hadoop related tools such as Pig and Hive for big data analytics
		CO4	Use Hadoop related tools such as Cassandra and HBase for big data analytics
		CO5	Learn data management using NoSQL
22HS351/301/401	**Business Economics and Financial Analysis**	CO1	Understand the concepts of micro and macro economics
		CO2	Analyse demand and forecast demand
		CO3	Evaluate markets and formulate competitive strategies
		CO4	Prepare financial statements
		CO5	Evaluate the financial strengths and weaknesses of a business by using ratio analysis
22DT353	**Fundamentals of Cloud Computing (PE-I)**	CO1	Explore Cloud Computing Environment
		CO2	Understand Parallel and Distributing Computing Architectures
		CO3	Use of Virtualization in Cloud Computing
		CO4	Understand the types of Cloud and Cloud model
		CO5	Describe different Cloud Platforms
22DT354	**Data Warehousing and Business Intelligence (PE-I)**	CO1	Explore the architecture of data warehouse and OLAP operations.
		CO2	Learn fundamental concepts of BI and Analytics
		CO3	Apply the Key Performance indicators of BI
		CO4	Implement Advanced BI Tools
		CO5	Implement of BI Techniques
22DT355	**Data Stream Mining (PE-I)**	CO1	Understand the concepts of MOA stream mining
		CO2	Apply methods / algorithms to new data stream analysis problems
		CO3	Apply the prediction strategies of Hoeffding Tree
		CO4	Evaluate the results and understand the functioning of the methods studied
		CO5	Demonstrate decision tree and adaptive Hoeffding Tree concepts
22AM354	**Soft Computing**	CO1	Learn about Soft Computing Techniques such as Neural Networks and Rough Sets
		CO2	Associative Memory and Adaptive Resonance Theory Analyze
		CO3	Use Fuzzy Set Theory and develop Fuzzy systems and applications
		CO4	Understand and apply Genetic Algorithms and Modeling for finding solutions real life Problems
		CO5	Understand Hybrid systems that use combination of soft computing techniques
22DT381	**Machine Learning and Predictive Analytics Lab**	CO1	Understand complexity of Machine Learning algorithms and their limitations
		CO2	Apply common Machine Learning algorithms in practice and implementing their own

		CO3	Perform experiments in Machine Learning using real-world Data
		CO4	Construct different predictive models for decision-making
		CO5	Apply descriptive modelling techniques for the given data
22DT382/434	**Big Data Analytics Lab**	CO1	Install Hadoop and perform basic file management task
		CO2	Implement basic data structures in Hadoop
		CO3	Apply map reduce technique to solve various problems
		CO4	Install Pig and Hive and execute to do basic programs
		CO5	Install MongoDB and do basic programs
22HS381/331	**Advanced English Communication & Soft Skills Lab**	CO1	Gain proficiency in communication skills
		CO2	Emerge as rational speakers
		CO3	Efficiently Manage their professional career
		CO4	Acclimatize to diverse cultures
		CO5	Be empowered with skills required for self-management
22DT383	**Industrial Oriented Mini Project**	CO1	Identify relevant, real-world problems and define clear, achievable project goals.
		CO2	Apply appropriate tools, models, and technologies to build end-to-end solutions.
		CO3	Clean, analyze, and visualize data to extract meaningful insights and patterns.
		CO4	Collaborate in teams with ethical responsibility and communicate outcomes effectively.
		CO5	Prepare detailed reports, manage project timelines, and present solutions confidently.

VII Semester			
Course Code	**Course Title**	**CO Number**	**Course Outcome Statement**
22DT401	**Web and Social Media Analytics**	CO1	Knowledge on decision support systems
		CO2	Apply natural language processing concepts on text analytics
		CO3	Understand sentiment analysis
		CO4	Discuss on search engine optimization and web analytics
		CO5	Optimise the multi-criteria system on Social Network analytics
22HS404/451	**Organizational Behaviour**	CO1	Understand the nature of Organizational Behavior and its applications in an organization
		CO2	Motivate themselves and others with a positive approach
		CO3	Work as a team member and build a good team
		CO4	Be good leaders
		CO5	Work effectively in an organization
22DT402	**Software Project Management (PE-II)**	CO1	Explore the evaluation of software process management
		CO2	Improve the software process architecture
		CO3	Disseminate life cycle and workflow of the process
		CO4	Analyse the minor and major milestones for process
		CO5	Design the project with modern tools
22DT403	**Spatial and Multimedia Databases (PE-II)**	CO1	Introduce the spatial data and models
		CO2	Representation of image database with R-Tree
		CO3	Apply image retrieval techniques
		CO4	Organizing video and audio data

		CO5	Optimize the multimedia data
22DT405	**Exploratory Data Analysis (PE-II)**	CO1	Handle missing data in the real-world data sets by choosing appropriate methods
		CO2	Identify the outliers if any in the data set
		CO3	Summarize the data using basic statistics. Visualize the data using basic graphs and Plots
		CO4	Choose appropriate feature selection and dimensionality reduction
		CO5	Apply techniques for handling high dimensional data
22AM409	**Introduction to Deep Learning**	CO1	Identify the various requirements for training a neural network
		CO2	Select appropriate parameters and architectures in building a deep network
		CO3	Choose Network Architecture, Vectorization and Normalization technique for real time data
		CO4	Examine different CNN Models for image and object recognition and classification
		CO5	Make use of RNN architectures to handle sequential or time series data
22IT403	**Quantum Computing (PE-III)**	CO1	Understand basics of quantum computing
		CO2	Understand physical implementation of Qubit
		CO3	Understand Quantum algorithms and their implementation
		CO4	Understand the Impact of Quantum Algorithms and its significance
		CO5	Realize the importance of current Asymmetric Algorithms on Quantum computing
22DT406	**NoSQL Databases (PE-III)**	CO1	Acquire the knowledge of NoSQL databases
		CO2	Articulate the distributed models
		CO3	Construct the database using MapReduce model
		CO4	Illustrate the document databases
		CO5	Distinguish the graph databases with respect to social networks
22AM405	**Computer Vision (PE-III)**	CO1	Understand the fundamental problems of computer vision
		CO2	Implement various techniques and algorithms used in computer vision
		CO3	Analyze and evaluate critically the building and integration of computer vision algorithms
		CO4	Demonstrate awareness of the current key research issues in computer vision
		CO5	Understand motion analysis using different analysis methods
22DT407	**Data Science Applications (PE-III)**	CO1	Correlate data science and solutions to modern problems
		CO2	Explore the data with time series and indexing
		CO3	Application of Natural Language Processing
		CO4	Describe the recommended systems
		CO5	Use of Genetic Algorithms
22DT431	**Web and Social Media Analytics Lab**	CO1	Knowledge on decision support systems
		CO2	Apply natural language processing concepts on text analytics
		CO3	Understand sentiment analysis
		CO4	Knowledge on search engine optimization and web analytics

		CO5	Optimise the multi-criteria system on Social Network analytics
22DT432	**Advanced Big Data Lab**	CO1	Write programming using Scala language.
		CO2	Develop applications using SparkFrame and API
		CO3	Create applications using Spark SQL and experiment spark streaming.
		CO4	Apply Machine Learning Algorithms in Spark for Big data Analytics
		CO5	
22DT433	**Project Stage - I**	CO1	Identify a real-world or research problem relevant to the domain.
		CO2	Perform a comprehensive literature survey and define project objectives.
		CO3	Conduct requirement analysis and define system specifications.
		CO4	Design system architecture and detailed module-wise components.
		CO5	Present project proposal, design documentation, and review progress effectively.

VIII Semester			
Course Code	**Course Title**	**CO Number**	**Course Outcome Statement**
22CY454	**Database Security (PE-IV)**	CO1	Ability to carry out a risk analysis for large database
		CO2	Ability to set up, and maintain the accounts with privileges and roles
		CO3	Gaining knowledge on security software design, database protection & intrusion detection systems
		CO4	Understanding different protection models related to databases like Object oriented, Relational etc.,
		CO5	Ability to understand the protection of new generation databases
22DT451	**Privacy Preserving in Data Mining (PE-IV)**	CO1	Understand the concepts of Privacy Preserving Data Mining Models and Algorithms.
		CO2	Explore Inference Control Methods for Privacy-Preserving Data Mining.
		CO3	Computation of Data Anonymization Methods and its Measures.
		CO4	Evaluate the solution designed for Multiplicative Perturbation.
		CO5	Implement the solutions for Utility-based Privacy Preserving Data
22DT452	**Social Network Analysis (PE-IV)**	CO1	Ability to construct social network maps.
		CO2	Use of NodeXL to perform social network analysis.
		CO3	Perform case studies on email, thread networks and Twitter.
		CO4	Gain skills in tracking the content flow through the social media.
		CO5	Analyse the content on YouTube.
22DT453	**Logistics and Supply Chain Management (PE-IV)**	CO1	Describing logistics and customer service
		CO2	Measuring the logistics cost
		CO3	Understand the importance of supply chain management in logistics

		CO4	Apply the knowledge of designing of network for SCM
		CO5	Solve demand and supply forecasting problems
22IT451	**Design Patterns (PE-V)**	CO1	Appreciate the basic concepts of design patterns and able to know how to select and use the design patterns
		CO2	Identify the design pattern in the existing code and use of creational patterns
		CO3	Apply and use the structural patterns
		CO4	Identify and use the behavioral patterns
		CO5	Find and catalog patterns in the object-oriented software
22IT453	**Information Retrieval Systems (PE-V)**	CO1	Understand the data structures like Inverted Indices used in Information retrieval systems.
		CO2	Understand the basics of web search.
		CO3	Understand the different techniques for compression of an index including the dictionary and its posting list.
		CO4	Understand various text search algorithms.
		CO5	Understand the evaluation metrics of information system.
22DT454	**Mining Massive Data Sets (PE-V)**	CO1	Handle massive data using MapReduce
		CO2	Develop the algorithms for massive data sets and methodologies in the context of data mining
		CO3	Understand the algorithms for extracting models and information from large Datasets
		CO4	Develop recommendation systems
		CO5	Gain experience in matching various algorithms for graph problems
22DT455	**Information Storage Management (PE-V)**	CO1	Understand the logical and physical components of a Storage infrastructure
		CO2	Explore the various forms and types of Storage Virtualization
		CO3	Discuss the storage architectures, including storage subsystems, DAS, SAN, NAS and CAS
		CO4	Describe the different roles in providing disaster recovery and business continuity capabilities
		CO5	Distinguish different remote replication technologies